

WOXSEN UNIVERSITY · ACADEMIC YEAR 2025-29

Event Registration and Venue Scheduling System

Database Management Systems — PBL Project

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PROBLEM STATEMENT

Scattered event data makes campus events hard to run

Scattered records
Event information scattered across different records.

Manual management
Organizers, categories, venues and schedules managed manually.

Registration tracking
No centralized participant registration tracking.

Inconsistent information
Duplicate registrations and inconsistent information.

Venue coordination
Venue allocation and timing need coordination.

Admin access
Admins lack quick access to registration, payment and event statistics.

A centralized relational database can integrate all these operations into one system.

One centralized database and web interface for every event

Capabilities

- Create and manage events; manage organizers and categories
- Manage venues and schedules; register participants
- Track registration and payment status; monitor event capacity; generate analytical reports

Objectives

- Centralize event data
- Model with 7 normalized entities
- Enforce integrity with constraints
- Prevent duplicate registrations
- Coordinate venues and timings
- Deliver analytics

WORKFLOW

Organizer → Event → Schedule/Venue → Student → Registration → Database → Reports

Technology Stack & Architecture

- Presentation Layer

User interface and forms

HTML · CSS · JavaScript

- Application/Backend Layer

PHP business logic and SQL

PHP

- Database Layer

MariaDB stores and enforces data

MariaDB/MySQL · phpMyAdmin

Server

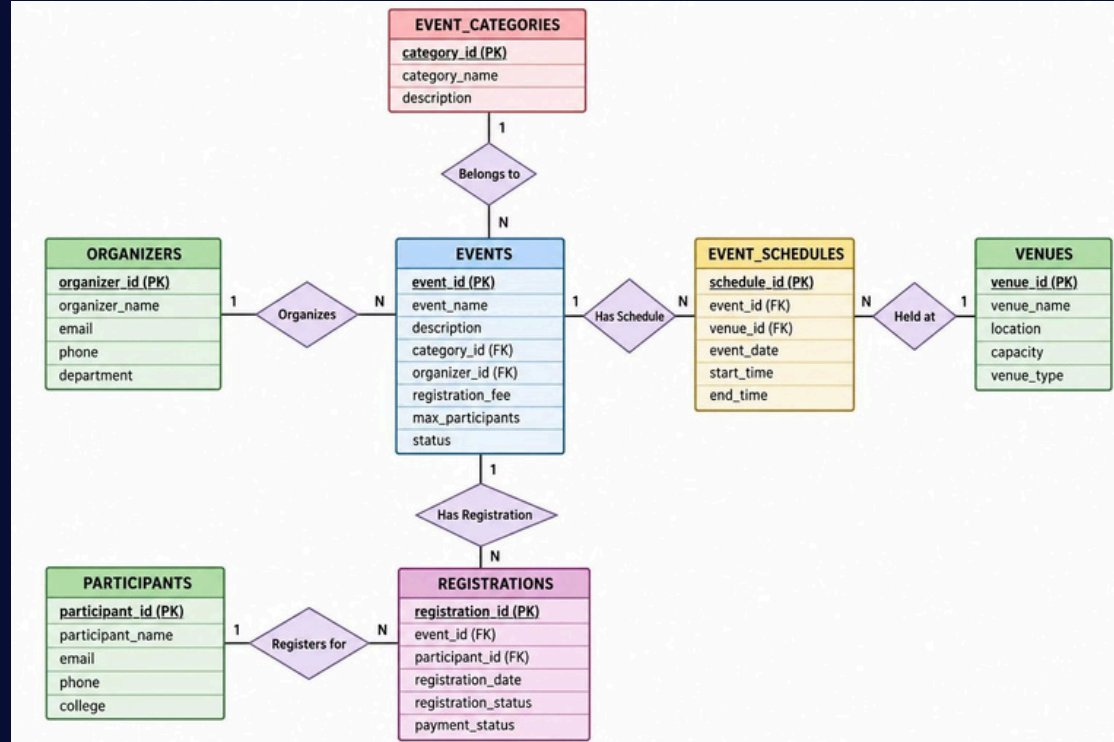
Apache/XAMPP

Development

VS Code

Git/GitHub

Entity Relationship Model



7 main entities

EVENT_CATEGORIES, ORGANIZERS, EVENTS

EVENT_SCHEDULES, VENUES

PARTICIPANTS, REGISTRATIONS

1:N relationships

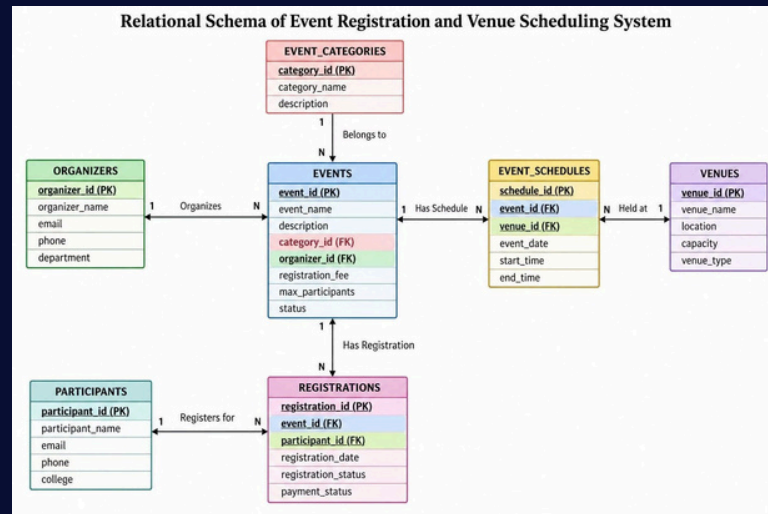
Categories → Events; Organizers → Events

Events → Schedules; Venues → Schedules

Events → Registrations; Participants → Registrations

PARTICIPANTS ↔ **EVENTS** is M:N through REGISTRATIONS.

Relational Schema & 3NF



EVENT_CATEGORIES (category_id PK, category_name, description)

ORGANIZERS (organizer_id PK, organizer_name, email, phone, department)

VENUES (venue_id PK, venue_name, location, capacity, venue_type)

EVENTS (event_id PK, event_name, description, category_id FK, organizer_id FK, registration_fee, max_participants, status)

EVENT_SCHEDULES (schedule_id PK, event_id FK, venue_id FK, event_date, start_time, end_time)

PARTICIPANTS (participant_id PK, participant_name, email, phone, college)

REGISTRATIONS (registration_id PK, event_id FK, participant_id FK, registration_date, registration_status, payment_status)

1NF

Atomic values; no repeating groups.

2NF

No partial dependency.

3NF

No transitive dependency.

Normalization reduces data redundancy and update anomalies.

Database Implementation

Built in MariaDB/MySQL through phpMyAdmin.

01	02	03	04	05	06
Create Database	Create Tables	Define PK/FK	Insert Data	Apply Constraints	Execute Queries

```
CREATE TABLE event_categories (category_id INT AUTO_INCREMENT PRIMARY KEY, category_name  
VARCHAR(100) NOT NULL UNIQUE, description VARCHAR(255));  
INSERT INTO events (...) VALUES (...);  
SELECT e.event_name, c.category_name FROM events e JOIN event_categories c ON e.category_id =  
c.category_id;
```

Constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT

System Modules & User Interface

The database connects to a local PHP web application with CRUD operations and real-time, database-driven information.



- Dashboard & Analytics
- Events Management
- Participant Management
- Registration Management
- Venues & Schedules
- Clubs/Organizers & Categories
- Student Portal
- SQL Query Console

Dashboard & Analytics

Project screenshot placeholder

Registration Management

Project screenshot placeholder

QUERIES AND OUTPUTS

SQL Queries & System Results

Queries

- Multi-table JOIN
- Registration count with COUNT()
- Organizer analytics; revenue with SUM()
- Venue capacity utilization
- GROUP BY and HAVING
- Above-average events via subquery
- Participant demographics

Demonstrates JOIN, aggregation, subqueries, GROUP BY, HAVING and conditional logic.



Dashboard / SQL Console

Project screenshot placeholder

Dashboard metrics

Total Events · Total Registrations · Enrolled Participants ·
Configured Venues · Revenue Collected · Category Distribution ·
Registration & Payment Metrics

Testing, Conclusion & Future Scope

Tested

Event creation, participant registration, venue scheduling, CRUD operations, SQL queries.

Validated

Foreign key relationships, duplicate registration prevention, UI/database consistency.

Conclusion

Requirements → ER Model → Relational Schema → 3NF → SQL Implementation → Application Integration → Testing

Future scope

Role-based authentication; automatic venue conflict detection; email/SMS notifications; waitlists and cancellations; attendance tracking; advanced analytics; PDF/CSV reports; production deployment.

Thank You